

cse 242 hw2

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1. (a)

$$\begin{aligned}
 J(w) &= \left| Xw - y \right|_2^2 + \lambda \left| w \right|_2^2 \\
 J(w) &= (Xw - y)^T (Xw - y) + \lambda w^T w \\
 J(w) &= w^T X^T X w - w^T X^T y - y^T X w - y^T y + \lambda w^T w \quad (1) \\
 J(w) &= w^T X^T X w - 2y^T X w + \lambda w^T w \\
 \frac{\partial J(w)}{\partial w} &= 2X^T X w - 2y^T X + 2\lambda w = 0
 \end{aligned}$$

then we see

$$\begin{aligned}
 2X^T X w + 2\lambda w &= 2y^T X \\
 (X^T X + \lambda I)w &= y^T X \quad (2)
 \end{aligned}$$

then we get the solution in the hw.

(b) as $\lambda \rightarrow 0$, $w_{ridge} \rightarrow (X^T X)^{-1} X^T y$. as $\lambda \rightarrow \infty$, $w_{ridge} \rightarrow 0$.
the former is when penalty goes to zero and the latter is when error dominates.

(c)

$$\begin{aligned}
 \left| \hat{X}w - \hat{y} \right| &= (\hat{X}w - \hat{y})^T (\hat{X}w - \hat{y}) \\
 &= w^T \hat{X}^T \hat{X} w - \hat{y}^T \hat{X} w - w^T \hat{X}^T \hat{y} + \left| \hat{y} \right|^2 \\
 &= w^T (X^T X + \sqrt{\lambda} X^T I_d + \sqrt{\lambda} I_d^T X + \lambda I) w \\
 &\quad - \hat{y}^T \hat{X} w - w^T \hat{X}^T \hat{y} + \left| \hat{y} \right|^2 \quad (3)
 \end{aligned}$$

note $(\sqrt{\lambda} I w)^2 = \lambda \sum w_i^2$. factoring out w^T , excluding the $w = 0$ solution, the argmin when zero gives the original expression.

2

(a)

$$\begin{aligned}
(X^T X + \lambda I)^{-1} &= (Q \Lambda^2 Q^T + \lambda I)^{-1} \\
&= (Q \Lambda Q^T + \lambda Q I Q^T)^{-1} \\
&= (Q(\Lambda^2 + \lambda I)Q^T)^{-1} \\
&= (Q^T)^{-1}((\Lambda^2 + \lambda I))^{-1}Q^{-1} \\
&= Q((\Lambda^2 + \lambda I))^{-1}Q^T
\end{aligned} \tag{4}$$

λ inversely scales the eigenvalue

(b)

$$\kappa(X^T X) = \frac{\sigma_{max}^2}{\sigma_{min}^2} \tag{5}$$

since sigmas are positive, the inequality on the right holds.

the equality on the left is true because of part a. the eigenvalues are exactly $\Lambda + \lambda I$

we can also see why equality only holds when λ is zero, since every eigenvalue is different on the LHS when λ is nonzero.

(c) if there are zero eigenvalues, it means the matrix is non invertable. the smallest eigenvalues with regularization are λ and no eigenvalues are zero, which implies full rank, which implies invertability.

2. (a) for ols

$$(X^T X)^{-1} X^T y = X^T y \tag{6}$$

for ridge

$$(X^T X + \lambda I)^{-1} X^T y = ((1 + \lambda)I)^{-1} X^T y = \frac{1}{1 + \lambda} X^T y \tag{7}$$

(b) using the hat symbol to denote estimate

$$\begin{aligned}
E[\hat{w}_{ridge}] &= E\left[\frac{1}{1 + \lambda} X^T y\right] \\
&= \frac{1}{1 + \lambda} X^T E[y]
\end{aligned} \tag{8}$$

$$E[y] = E[Xw^* + \epsilon] = Xw^* + E[\epsilon] = Xw^*$$

then

$$E[\hat{w}_{ridge}] = \frac{w^*}{1 + \lambda}$$

similarly

$$\begin{aligned}
Var(\hat{w}_{ridge}) &= Var\left(\frac{1}{1+\lambda}X^Ty\right) \\
&= \left(\frac{1}{1+\lambda}\right)^2 X^T Var(y)(X^T)^T \\
&= \frac{1}{(1+\lambda)^2} X^T Var(y) X \\
&= \frac{\sigma^2}{(1+\lambda)^2} X^T X
\end{aligned} \tag{9}$$

(c)

$$\begin{aligned}
Bias_j &= E[w_{ridge,j}] - w_j^* \\
&= \frac{w_j^*}{1+\lambda} - w_j^* \\
&= w_j^* \left(\frac{1}{1+\lambda} - 1\right) \\
&= w_j^* \left(\frac{1 - (1+\lambda)}{1+\lambda}\right) \\
&= \frac{-\lambda w_j^*}{1+\lambda}
\end{aligned} \tag{10}$$

therefore

$$MSE_j = \left(\frac{\lambda w_j^*}{1+\lambda}\right)^2 + \frac{\sigma^2}{(1+\lambda)^2}$$

to find optimal λ , w wolfram alpha

$$\frac{d}{d\lambda} MSE_j = \frac{2(\lambda(w_j^*)^2 - \sigma^2)}{(1+\lambda)^3}$$

set λ to $\frac{\sigma^2}{(w_j^*)^2}$ to make the numerator 0 and thus the whole term 0.

OLS is bad when noise is high and good when noise is low. ridge is better when noise is high.

3. (a)

$$P(w|X, y) = \frac{P(y|X, w)P(w)}{P(y|X)}$$

$$\ln P(w|X, y) = \ln P(y|X, w) + \ln P(w) - \ln P(y|X)$$

both terms are normal distributions so we see

$$\ln P(y|X, w) = -\frac{1}{2\sigma^2} \|y - Xw\|_2^2 + C_1$$

$$\ln P(w) = -\frac{1}{2\tau^2} \|w\|_2^2 + C_2$$

note, after taking log the constant normalization terms for P become additive constants.

thus

$$\ln P(w|X, y) = -\frac{1}{2\sigma^2} \|y - Xw\|_2^2 - \frac{1}{2\tau^2} \|w\|_2^2 + C$$

but this is just the negative of the ridge objective up to constants so maximizing this is the same as minimizing the ridge constant obj.

- (b) if λ is large, τ^2 is small since $\lambda = \left(\frac{\sigma}{\tau}\right)^2$. the gaussians have small variance and are near zero. since λ controls the weight given to the prior by influencing τ , it exhibits similar behavior.

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1. (a)

$$\begin{aligned} J(w) &= (Xw - y)^T (Xw - y) + \lambda w^T w \\ &= (w^T X^T X w - w^T X^T y - y^T X w + y^T y) + \lambda w^T w \quad (11) \\ &= (w^T X^T X w - 2w^T X^T y + y^T y) + \lambda w^T w \end{aligned}$$

$$\nabla_w J(w) = 2X^T X w - 2X^T y + 2\lambda w \quad (12)$$

2. see jupyter notebook
3. 0.001 converges the fastest. 0.01 diverges. 0.001 goes fast enough but does not overshoot. 0.01 overshoots too much.
4. see jupyter notebook
they are not that close.